

|          |                                      |           |
|----------|--------------------------------------|-----------|
|          |                                      |           |
| 6(a)(i)  | $1/T = 1/(2C) + 1/C$                 | <b>C1</b> |
|          | $T = \frac{2}{3}C$ <b>or</b> $0.67C$ | <b>A1</b> |
| 6(a)(ii) | same charge on Q as on combination   | <b>B1</b> |
|          | so p.d. is 6.0 V                     | <b>B1</b> |
| 6(b)     | P: p.d. will decrease (from 3.0 V)   | <b>B1</b> |
|          | to zero                              | <b>B1</b> |
|          | Q: p.d. will increase (from 6.0 V)   | <b>B1</b> |
|          | to 9.0 V                             | <b>B1</b> |